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Understanding the Impacts of Training LLMs on CPU and GPU Architectures

1. Abstract

Large Language Models (LLMs) have pushed computing hardware to its limits. While GPUs dominate large-scale training due to their multi-core parallel throughput, CPUs remain critical for preprocessing, orchestration, and small-scale workloads. Yet, the architectural trade-offs between these processors for LLM training are not well understood. This project aims to characterize the performance, efficiency, and bottlenecks of LLM training across CPU and GPU architectures using small transformer workloads. By profiling runtime phases such as data transfer, forward pass, backward pass, and optimization, the study will identify where GPUs outperform CPUs and when CPU training remains competitive. Using real hardware and profiling tools, this work will clarify how hardware design influences LLM efficiency and provide insight for future AI system optimization.

2. Introduction and Motivation

As model sizes and dataset scales expand, efficient training has become one of the central challenges in computer architecture. GPUs excel at parallelized tensor operations, but they face limits in memory capacity, energy consumption, and data-transfer overhead. CPUs, while slower in raw throughput, often achieve better performance on smaller or memory-constrained models. Understanding how training workloads map to hardware behavior is key to optimizing both cost and energy efficiency.

This project investigates how CPU and GPU architectures handle the computational and memory demands of LLM training. By empirically measuring performance rather than simulating it, the goal is to expose real bottlenecks and evaluate how architectural factors—such as parallelism, bandwidth, and data movement—affect efficiency. These findings will inform strategies for balancing workloads across heterogeneous systems.

3. Problem Statement and Importance

Training LLMs involves intensive matrix multiplications, memory accesses, and inter-device communication. GPUs provide exceptional throughput but rely on high data-transfer bandwidth between CPU and GPU memory. CPUs, by contrast, have lower parallel density but higher single-thread efficiency and easier data access. The key question is: *At what workload scale does GPU training outperform CPU training, and why?* Answering this has practical importance for AI research labs and companies working on smaller transformer models that cannot fully utilize large GPU clusters. The study emphasizes hardware-level inference rather than simulator-based evaluation, aligning with recent efforts to analyze machine learning workloads directly on physical systems.

4. Key Insight and Hypothesis

For small transformer models and low batch sizes, CPU-based training can outperform GPU execution due to reduced data-transfer overhead and higher per-core memory efficiency. As model and batch size increase, GPUs surpass CPUs once their parallel units become fully utilized. By breaking down each phase of training, this study aims to locate the performance crossover point and identify which hardware subsystems—memory, compute, or interconnect—dominate training cost.

5. Methodology

The experiment will use PyTorch to train a lightweight transformer model under identical configurations on both CPU and GPU. The training phases—data transfer, forward pass, backward pass, and optimization—will be timed. Also important metrics such as execution time, memory usage, and compute utilization will be collected across varying models and batch sizes in order to analyze scaling behaviors. The metrics collected will then be plotted to reveal bottlenecks and identify the performance crossover.

Hardware and Tools:

- Hardware: PC with multicore CPU and GPU (preferably NVIDIA)
- Tools: Pytorch for model, NVIDIA Nsight Systems for GPU kernel and memory analysis, and Intel VTune for CPU performance counters

6. Metrics of Evaluation

- Execution Time: Total training and per-stage breakdown
- Throughput: Samples processed per second
- Memory Utilization: DRAM and VRAM usage efficiency
- Compute Utilization: Active core or kernel usage percentage

7. Related Work

LLM inference and performance on hardware has been researched in recent years due to the sudden growth in machine learning. Hengrui Zhang (2023) introduced a hardware evaluation framework for LLM workloads, establishing cross-platform baselines for CPUs, GPUs, and TPUs and empirical performance comparison. Jiazhi Jiang (2022) analyzed transformer inference on multi core CPUs, revealing that memory bandwidth and thread scheduling dominate performance—insights that extend naturally to training workloads. Most recently, Haolin Zhang (2025) demonstrated that CPUs can outperform GPUs for on-device LLM inference under certain data transfer and batch size conditions, reinforcing the hypothesis that smaller models may favor CPU architectures. Building on these studies, this project focuses specifically on training-phase bottlenecks and hardware efficiency crossover between CPU and GPU systems.

Work Cited

Jiang, Jiazhi. "Characterizing and Optimizing Transformer Inference on Arm Many-Core Processor | Proceedings of the 51st International Conference on Parallel Processing."

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